Experimental Investigation of Wind Turbine Performance:
Energy Balance and Efficiency of HAWT and VAWT Rotors

Wind Turbine Performance Study

Energy conversion, efficiency and wake behaviour of HAWT and Savonius VAWT rotors



Armfield RE12 Advanced Wind Energy System

BEng (Hons) Mechanical Engineering

London South Bank University | Year 2

2026 • Practical experimental investigation

What I set out to investigate

A controlled laboratory comparison of two small-scale wind turbine rotor designs.

AIM

Investigate how wind energy is converted into electrical output and compare the performance of a three-blade HAWT with a Savonius VAWT.

MEASUREMENTS

Upstream and downstream air velocity, generator terminal voltage and current across eight fan supply settings (5–12 V).

ENGINEERING QUESTIONS

How does wind speed affect available power? How do rotor design and starting behaviour affect electrical output and efficiency?

Key context: 23.0 °C ambient temperature • 65.5% relative humidity • ρ = 1.1731 kg/m^3 • 33 Ω load

The Armfield RE12 test rig

Photographs taken during the practical session.



WIND GENERATOR

Axial fan and flow-straightening section produce a controlled airflow.

POWER & LOAD MODULES

Variable DC supply, generator circuit and fixed 33 Ω load.

MEASUREMENT

Anemometer for air velocity; electrical module for voltage and current.

Two different approaches to extracting wind energy

Both rotors were tested on the same RE12 platform.





Left: three-blade HAWT | Centre: Savonius VAWT

HAWT • LIFT-BASED

Three blades; 25° pitch; 62 mm blade-tip radius; swept area 0.012076 m^2.

SAVONIUS VAWT • DRAG-BASED

Vertical axis; 80.3 mm diameter × 60 mm height; projected area 0.004818 m^2.

From raw readings to performance metrics

The same measurement sequence was repeated for each rotor.

1. Set ambient conditions and measure rotor dimensions.
2. Calibrate upstream wind speed without a rotor fitted.
3. Fit the HAWT; record velocity, voltage and current at 5–12 V.
4. Replace with Savonius VAWT and repeat the measurements.
5. Calculate available wind power, electrical output and efficiency.

CALCULATION FRAMEWORK

Available wind power: $P_v = \frac{1}{2}\rho A V_v^3$

Electrical output: $P_r = UI$

Power coefficient: $C_p = P_r / P_v$

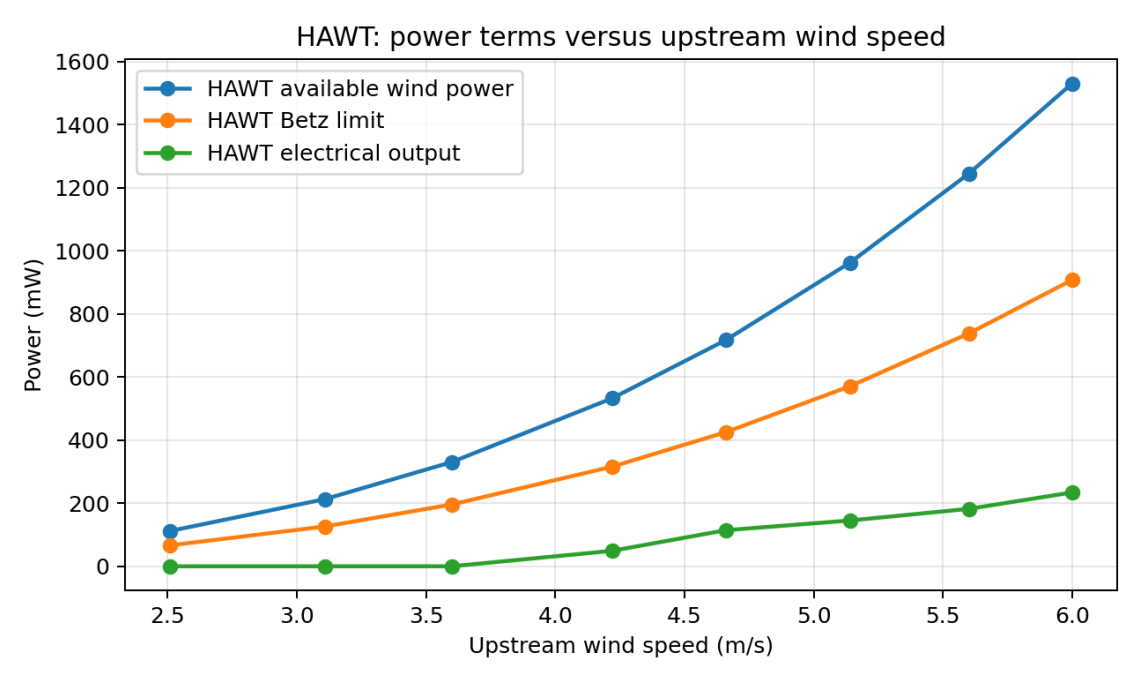
Betz-limit power: $P_\beta = 0.593 P_v$

Efficiency relative to Betz: $\eta = P_r / P_\beta \times 100\%$

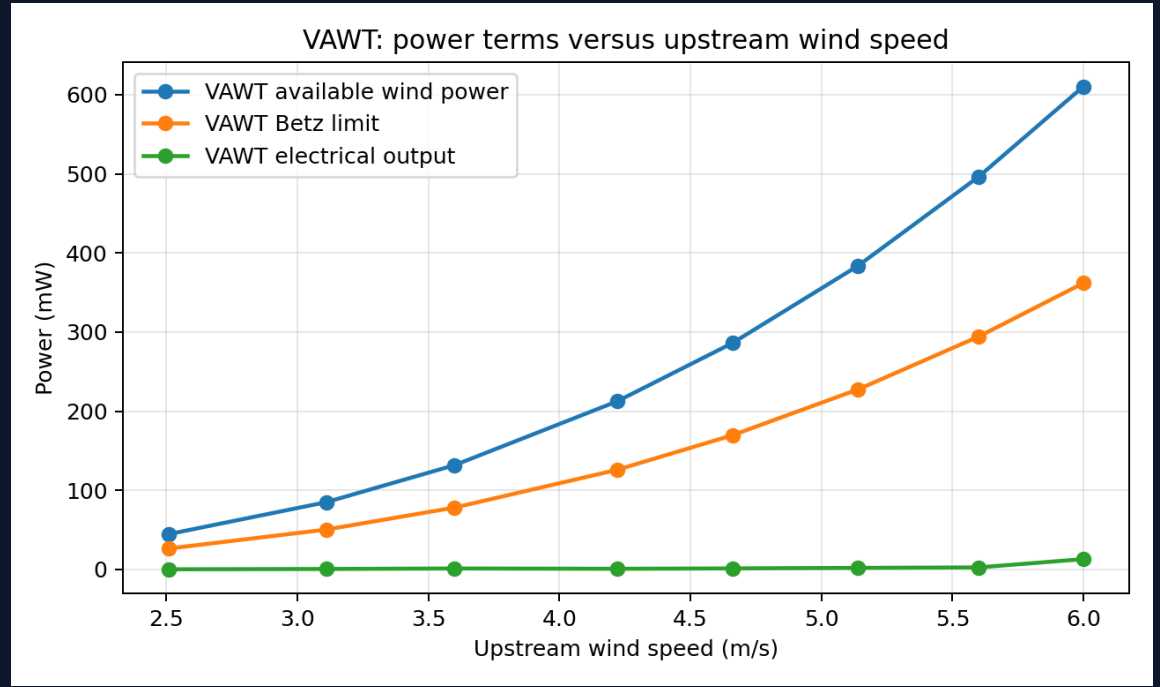
Downstream velocity was used to examine wake deceleration, not as a direct substitute for the standard upstream power calculation.

Measured performance trends

Electrical output and theoretical available power increased with upstream wind speed.



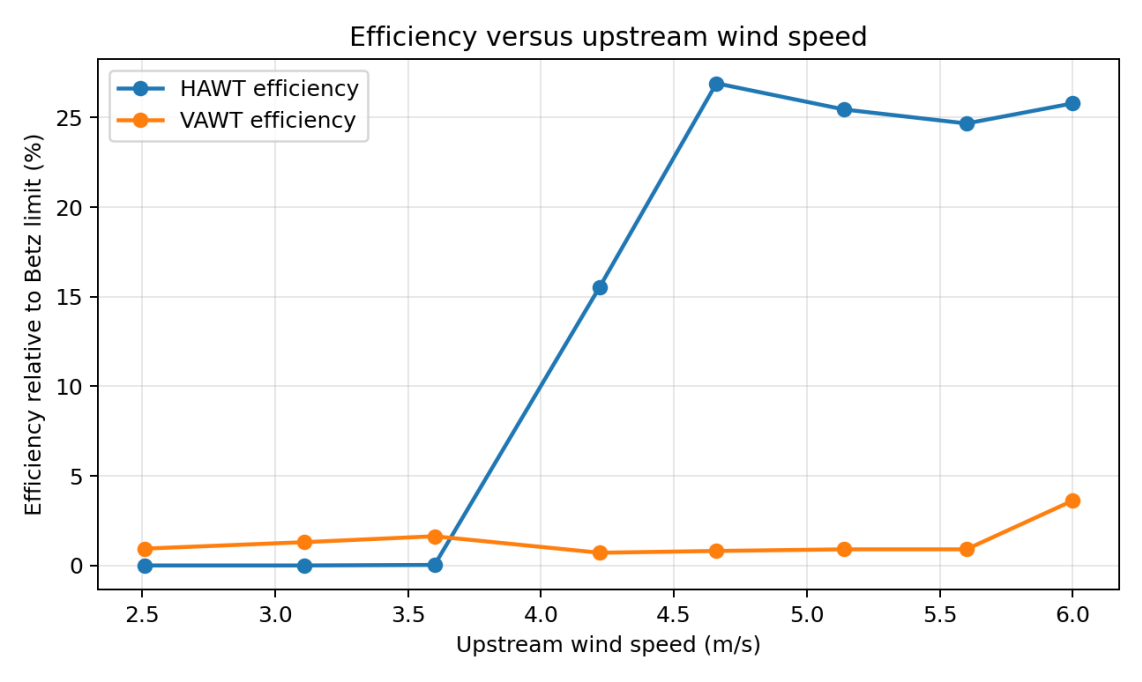
HAWT: peak recorded output 234 mW at 6.00 m/s. The rotor did not produce output at the two lowest test settings.

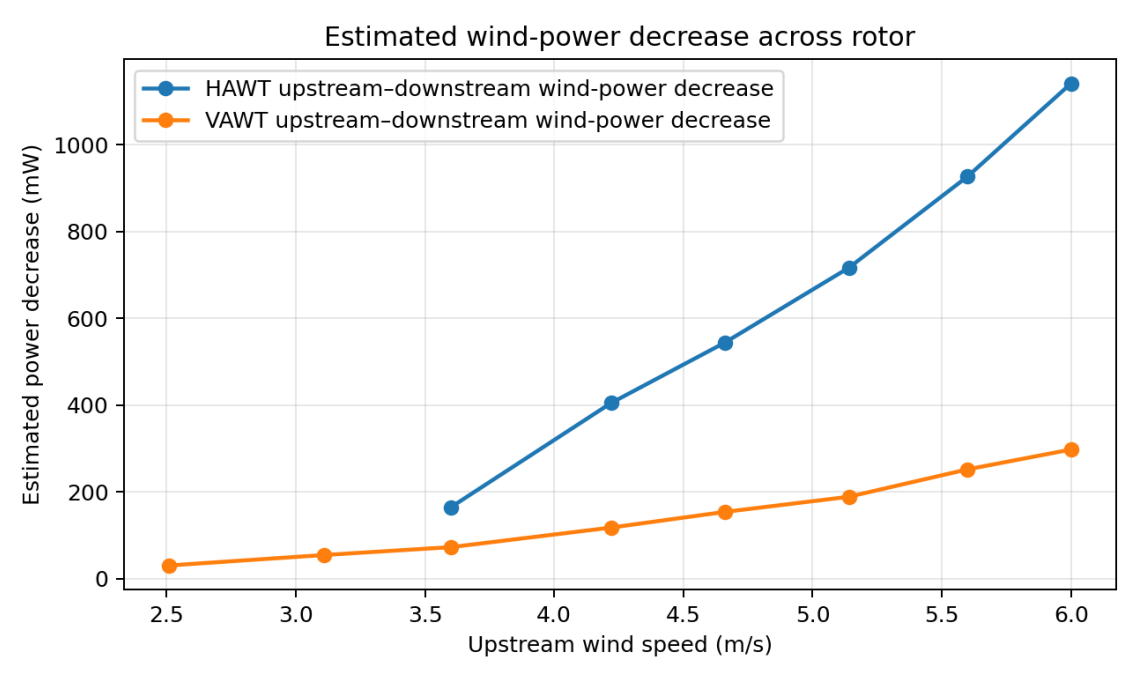


Savonius VAWT: peak recorded output 13.11 mW at 6.00 m/s; it self-started at the lowest tested wind speed.

Efficiency and downstream wake

The two rotor designs showed distinctly different behaviour under the same laboratory conditions.





HAWT

At 6.00 m/s: 25.79% efficiency relative to the Betz limit (reported value).

SAVONIUS VAWT

Peak efficiency relative to Betz limit: 3.62% (reported at 6.00 m/s).

What this practical taught me

Connecting thermofluids theory with measured engineering behaviour.

AERODYNAMICS

Rotor geometry and operating principle influence how much kinetic energy can be extracted. Lift-based and drag-based rotors behave differently.

ENERGY CONVERSION

The Betz limit provides a theoretical benchmark. Real output is lower because of aerodynamic, mechanical and electrical losses.

EXPERIMENTAL SKILLS

I practised setting up instrumentation, recording repeatable measurements, processing data, plotting performance curves and interpreting uncertainty.

Limitations to consider: small-scale rotor Reynolds number, flow bypass and wake turbulence, fixed blade pitch, bearing/generator losses and a fixed resistive load. Results are specific to this bench-scale rig.

Next step: apply experimental analysis and engineering problem-solving to placement projects in 2027.

Comprehensive experimental measurements

All eight excitation levels recorded across a fixed 33.0 Ω load resistor.

Raw Readings

HAWT • THREE-BLADED (A = 0.012076 M²)

Vfan	VW	PW(mW)	U(V)	I(mA)	PR(mW)	ηBetz(%)
12.0 V	**6.00**	**1530.00**	**1.56**	**150.0**	**234.00**	**25.79%**
11.0 V	5.60	1243.95	1.40	130.0	182.00	24.67%
10.0 V	5.14	961.89	1.23	118.0	145.14	25.45%
9.0 V	4.66	716.80	1.10	104.0	114.40	**26.91%**
8.0 V	4.22	532.32	0.70	70.0	49.00	15.52%
7.0 V	3.60	330.48	0.07	0.8	0.06	0.03%
6.0 V	3.11	213.07	0.00	0.0	0.00	0.00%
5.0 V	2.51	112.01	0.00	0.0	0.00	0.00%

SAVONIUS VAWT (A = 0.004818 M²)

Vfan	VW	PW(mW)	U(V)	I(mA)	PR(mW)	ηBetz(%)
12.0 V	**6.00**	**610.42**	**0.186**	**70.5**	**13.11**	**3.62%**
11.0 V	5.60	496.28	0.165	16.0	2.64	0.90%
10.0 V	5.14	383.75	0.147	13.9	2.04	0.90%
9.0 V	4.66	285.97	0.115	11.9	1.37	0.81%
8.0 V	4.22	212.37	0.096	9.3	0.89	0.71%
7.0 V	3.60	131.84	0.114	11.2	1.28	1.63%
6.0 V	3.11	84.99	0.082	8.0	0.66	1.30%
5.0 V	2.51	44.69	0.050	5.0	0.25	0.94%

Sample calculation at 12.0 V (HAWT): $P_W = 0.5 \times 1.1731 \times 0.012076 \times 6.00^3 = 1530.00$ mW | $P_R = 1.56$ V × 150.0 mA = 234.00 mW | $\eta_{Betz} = 234.00 / (0.593 \times 1530) =$ **25.79%**

Cubic power scaling and starting dynamics

Validation of aerodynamic scaling and cut-in torque thresholds.

Fluid Mechanics

SCALING VERIFICATION

$(2.39)^3 = 13.65\times$ Growth

Increasing excitation from 5.0 V to 12.0 V produced a 2.39-fold wind velocity rise (2.51 to 6.00 m/s). Available kinetic power grew by exactly **13.65-fold**, strictly conforming to the theoretical cubic velocity law ($P \propto V_W^3$).

Area ratio AHAWT / AVAWT = 0.012076 / 0.004818 = **2.506** (Available power HAWT is exactly 2.506× VAWT).

CUT-IN SPEED MECHANICS

3.11 – 3.60 m/s Threshold

The HAWT stalled completely at 5.0 V and 6.0 V because static bearing stiction and generator magnetic cogging torque exceeded aerodynamic starting lift. Once passing 3.60 m/s, attached lift formed rapidly, reaching 26.91% peak Betz efficiency.

Savonius VAWT self-started at 2.51 m/s (0.25 mW) due to constant positive static drag torque on concave cups.

Laboratory vs commercial utility turbines

Evaluating aerodynamic scale disparity and electromechanical constraints.

Reynolds Scale

SCALE & REYNOLDS NUMBER DISPARITY

Low Reynolds Regimes ($Re \approx 10^4$): Laboratory 62 mm blades suffer thick laminar boundary layers, premature separation, and low lift-to-drag ratios ($C_L/C_D < 15$).

Commercial Turbines ($Re > 10^6$): Slender, twisted airfoils maintain fully attached turbulent flow, reaching $C_p = 0.45–0.50$ (75–85% of Betz limit).

ELECTROMECHANICAL & CONTROL CONSTRAINTS

Fixed Resistance (33.0 Ω): Violated electrical impedance matching ($R_{int} = R_{load}$) across varying speeds, heavily loading the slow-turning Savonius generator.

Fixed 25° Pitch vs Active MPPT: Real utility systems use active pitch regulation and Maximum Power Point Tracking to lock optimal tip-speed ratios ($\lambda \approx 6–8$).

Thermodynamic energy balance

Entropy generation and exergy degradation mechanisms within the control volume.

Energy Balance

Wake Kinetic Deficit

Air continuity dictates air must exit the disc ($V_{down} > 0$). Betz theory proves an ideal exit velocity of $V_W/3$ (33.3%). HAWT wake dropped to 3.80 m/s (63.3%), losing kinetic exergy into downstream swirl.

$V_{down} / V_W = 0.633$

Tip Vortex & Bypass

Without tip winglets, high-pressure air leaks over blade tips into spanwise vortexes. Unconfined wind tunnel flow partially bypasses the high-impedance rotor disc entirely.

Induced Tip Drag

Friction & Joule Heat

Total energy is conserved (1st Law); however, available work is irreversibly degraded into low-grade thermal dissipation through mechanical bearing friction and generator I^2R winding heating.

$\Delta S_{gen} > 0$ (Exergy Loss)

Lift vs drag engineering comparison

Evaluating HAWT and Savonius VAWT across 5 critical engineering criteria.

Evaluation Matrix

ENGINEERING CRITERION	HAWT (3-BLADED LIFT ROTOR)	SAVONIUS VAWT (DRAG ROTOR)
1. Axis & Orientation	Horizontal axis; requires active yaw mechanism to track wind.	**Vertical axis; omnidirectional (360°) wind capture without yaw.**
2. Rotational Speed (TSR)	**High shaft speed; tip-speed ratio λ > 1; efficient generator coupling.**	Low shaft speed; bucket cannot exceed wind speed (λ < 1).
3. Starting Characteristics	Stalled below 3.60 m/s due to blade stall and bearing stiction.	**Self-starting down to 2.51 m/s via positive static torque.**
4. Voltage & Output	**Up to 1.56 V, 150.0 mA (234 mW peak power output).**	**Limited to 0.186 V, 70.5 mA (13.11 mW peak power output).**
5. Operating Principle	Bernoulli lift differentials generate driving aerodynamic torque.	Differential drag between concave cup and returning convex cup.

PLACEMENT CANDIDATE PROFILE • 2027 INDUSTRIAL PLACEMENT

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1. Experimental Rig Operation

Tunnel calibration, electrical load measurement, anemometer data logging.

2. MATLAB & Data Modeling

Automated curve generation, cubic power validation, Betz efficiency routines.

3. Thermofluids & Exergy Analysis

Actuator disc momentum models, wake deceleration, irreversible exergy tracking.

4. Technical Reporting & Synthesis

Transforming complex experimental datasets into structured executive insights.

Full Laboratory Report & MATLAB Code Available: S4431048@lsbu.ac.uk • London South Bank University

Seeking 2027 Industrial Placement